OligoTox-Coagulopathy

Coagulation toxicity of oligonucleotide therapeutics · Narrative document · NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2

213	2,388	941	75	55/55
oligonucleotides	coagulation measurements	per-position modification records	sources, every document held	QC checks pass

1. Executive summary

OligoTox-Coagulopathy pairs the design of 213 oligonucleotides with 2,388 measured coagulation outcomes drawn from 75 sources, every one of which is held in the release so that any row can be re-checked against the document it cites. 47 compounds carry a full per-nucleotide chemistry map, expanded into a 941-row position table; 97 of 213 (46%) carry a published sequence. Coagulopathy is the fifth endpoint on the Challenge's list of toxicities of interest, and before this release the public literature offered no structured, sequence-resolved account of it.

The endpoint is prolongation of clotting times, fibrinogen and factor disturbance, and the bleeding or thrombotic outcomes that follow from them. Platelet count alone is excluded: thrombocytopenia is a separate endpoint on the same list, and conflating the two is the commonest way this literature is misread.

Positive and negative controls

The dataset is not a list of toxic compounds. It spans the full range and carries controls at both ends, which is what makes it trainable:

- **Measured negatives — 572 rows.** An endpoint that was assessed and found unremarkable. These are distinguished, deliberately and throughout, from an endpoint that was never measured: the latter is `NOT_REPORTED` with the reason recorded, never `no_change`. Independent reviewers tested this distinction specifically and could not break it.
- **Vehicle, saline and untreated-plasma comparators** carried as the matched control on the row they belong to, in `control_value`, rather than as separate rows a user must find and join.
- **Severe positives — 26 rows at grade 3**, the highest severity the rubric assigns, plus 15 rows carrying a severity grade the source itself reported.
- **Pharmacological positive controls.** 1,720 rows measure compounds designed to alter coagulation — anti-factor-XI and anti-factor-XII antisense, prekallikrein and factor-VII programmes, anticoagulant aptamers, antithrombin-lowering siRNA. They are a mechanistically-explained positive class, and they are flagged so they are never mistaken for toxicity.

> **The single most important thing about this dataset.** 1,720 of 2,388 rows are **on-target pharmacology, not toxicity**, because the compounds with published clotting numbers are largely the ones designed to change clotting. Two boolean columns keep the axes apart and **both may be true on one row** (144 rows are). A model trained across them without the flags learns that anticoagulants prolong aPTT — true, circular, and useless for safety prediction.

2. Main findings and conclusions

Finding 1 — the phosphorothioate class effect, quantified per compound for the first time

The safety literature states this effect at class level: prolongation of coagulation time at high plasma C_{max} of phosphorothioate-backbone oligonucleotides, independent of sequence and of hybridisation. This release expresses it as per-compound numbers. Across rows that are *not* on-target pharmacology, full-phosphorothioate compounds show a median aPTT of **1.42× the matched control** (n = 48).

The caveat is load-bearing and the figure is drawn to show it: **42 of those 48 rows come from a single source**. It is one well-controlled experiment, not a meta-analysis, and it should not be cited as though it were.

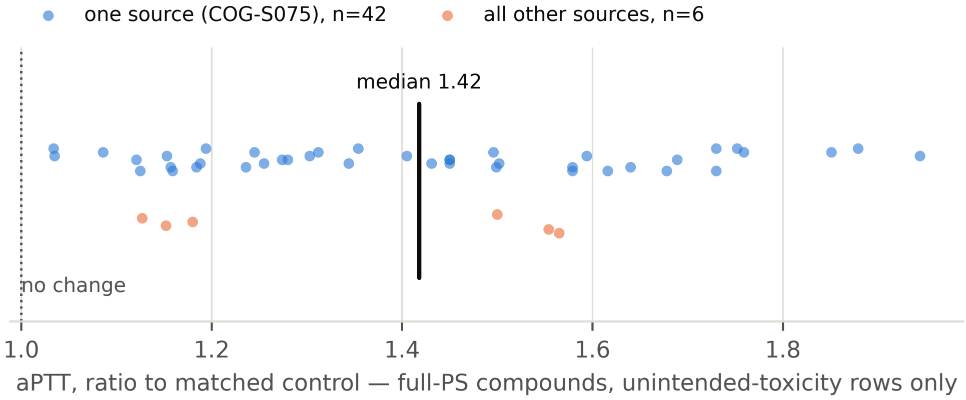

Figure 3. aPTT prolongation for full-phosphorothioate compounds, unintended-toxicity rows only. Points are split by source to make the concentration of evidence visible: 42 of 48 come from one study.

Finding 2 — a source whose own prose contradicts its own tables

US 9,061,044 states verbatim that "PT, aPTT and fibrinogen were not significantly altered in monkeys treated with ISIS oligonucleotides compared to the PBS control." Its Table 87 shows every treated group above control at every timepoint, in a clean compound rank order peaking at 4 h — 39.13 s against a 20.13 s control, a 1.94× prolongation. The dataset extracts the tables and carries the contradicting sentence verbatim on all 126 of that source's rows, so the disagreement travels with the data. Any pipeline that reads conclusions rather than tables records a false negative here.

Finding 3 — the standard in-vitro assay cannot do the job it is used for

One source demonstrates that in-vitro aPTT is saturated by phosphorothioate content and therefore cannot discriminate toxic from non-toxic compounds. Nulls from that assay are encoded as a *method limitation*, never as a safety finding. The opposite reading — the intuitive one — would teach a model that a saturated assay means a safe compound.

Finding 4 — human and animal evidence are separated, and rarely paired

886 measurements are made in human or human-derived systems and 1,476 in animal systems. Only **30 of 213 compounds carry both**, which bounds what any translation model built on this release can claim.

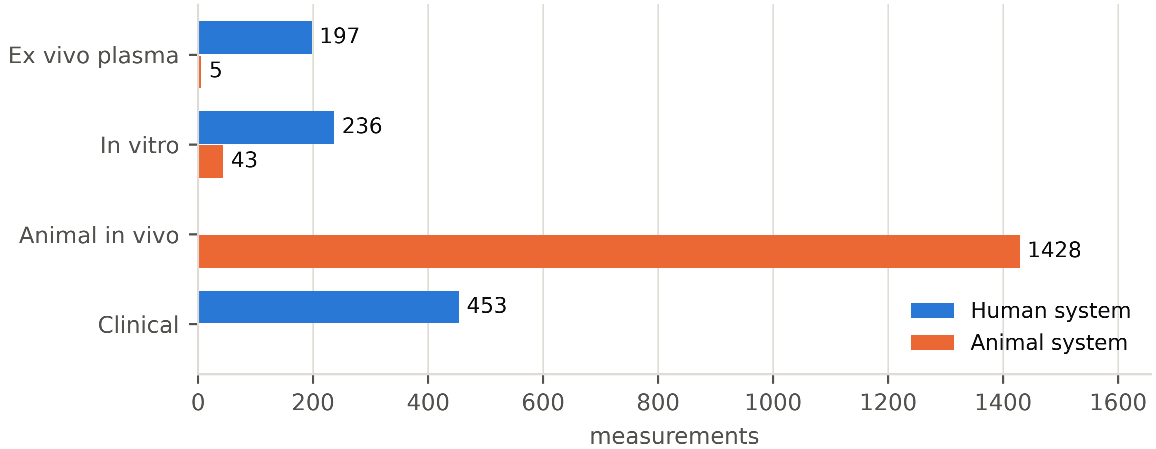

Figure 1. Human versus animal evidence by study design. The clinical and animal-in-vivo arms are disjoint by construction; the in-vitro and ex-vivo arms are where human systems and animal systems both appear.

3. How the data were produced

This is an **in-silico curation** of published data; no wet-lab experiment was performed. Sources were found by eight independent search axes — mechanism, clinical trials, nonclinical toxicology, siRNA and lipid nanoparticles, aptamers, regulatory labels, patents, and reviews used as a citation map — followed by a targeted sweep for human data. Every candidate had to be *retrieved and quoted* before entering the work-list; nothing rests on a search-engine summary. That discipline closed the project's only previously recorded lead for this endpoint, which proved on retrieval not to report the values its title implied.

Extraction rules, enforced by the output contract: every measurement carries a verbatim quote and an exact locus; no value is ever read off a figure (467 rows are therefore qualitative rather than invented); per-position chemistry is transcribed from the source's own legend, never modelled. Retrieval used the Europe PMC REST service, DailyMed, the ClinicalTrials.gov API, EMA and FDA review documents, and the USPTO full-text endpoint. The pipeline is deterministic and committed: from a clean checkout, `build_dataset.py → validate_dataset.py → verify_against_sources.py` reproduces and re-checks the release with no network access.

Computational processing, and what it corrects

The build applies corrections that an adversarial verification pass identified, rather than leaving them as hand edits: a "relative" clotting time is a subtracted delta and must not be divided by its control; percent *inhibition* is not percent *of control*; a combination arm must be referenced to the partner-drug arm, not to the untreated cell; 120 pre-dose baselines are not effect measurements; and a source-stated measured null outranks a ratio a hair above 1.00. 17 rows carry a `co_administered_agent` and are not measurements of the oligonucleotide alone.

4. Indicators, predictors and their distributions

Response variables — how coagulation toxicity was measured

Axis	Distribution
Readout category	clotting time 1160 · factor activity 387 · bleeding outcome 240 · thrombotic outcome 220 · fibrinogen 144 · platelet coag crosstalk 83 · anticoagulant activity 76 · thrombin generation 47 · fibrinolysis marker 31
Readout (top)	aPTT 599 · PT 376 · fibrinogen 140 · FXI_activity 108 · antithrombin_activity 75 · thrombus_platelet_accumulation_fluorescence 65 · template_bleeding_time 58 · fibrin_formation_fluorescence 52
Study design	clinical 453 · animal in vivo 1,433 · in vitro 297 · ex vivo plasma 202
Species	human 850 · monkey 818 · mouse 569 · rat 31 · pig 21 · minipig 17
System origin	human 886 · animal 1,476 · not determined 26

Grades are ordinal 0–3, assigned **mechanically** from a control-referenced ratio by published **CTCAE v5.0** cut-offs, and only for the readouts CTCAE defines. 1,521 rows (64%) are deliberately **left ungraded**, each stating why, rather than graded by a threshold invented here.

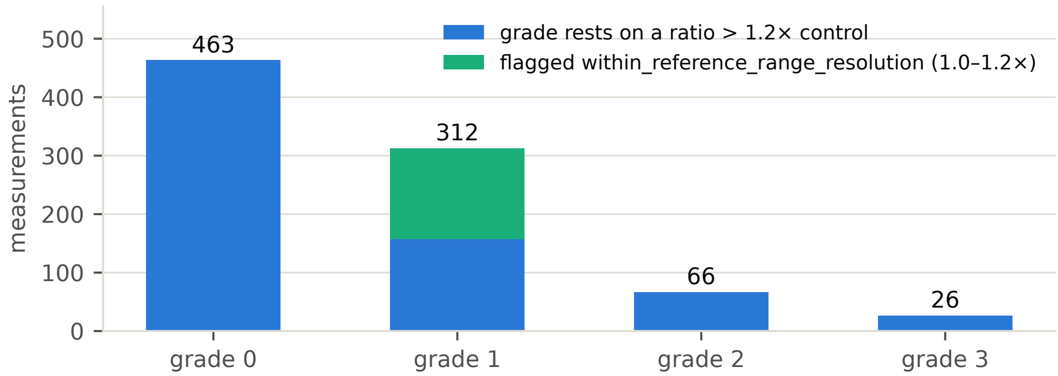

Figure 4. Grade distribution. The upper segment marks grades resting on a ratio between 1.0 and 1.2× control, which normal variation cannot be excluded from.

> **Read the grade column with its caveat.** CTCAE grades against the upper limit of normal; these sources publish a control mean. A ratio a few percent above 1.00 is therefore not evidence of a real prolongation. 155 graded rows carry `grade_caveat = within_reference_range_resolution`. Filter on it before treating grade 1 as a finding.

Predictor variables — the distribution across tested oligos

Variable	Distribution across 213 compounds
Class	aptamer 59 · ASO gapmer 58 · other 36 · GalNAc siRNA 28 · siRNA 13 · tcDNA ASO 7 · PMO 4
Backbone	NOT REPORTED 99 · full PS 57 · mixed PO PS 34 · full PO 14 · PMO neutral 4
Sequence published	97 / 213
Position-resolved chemistry	47 / 213 (941 position records)
Human *and* animal data	30 / 213

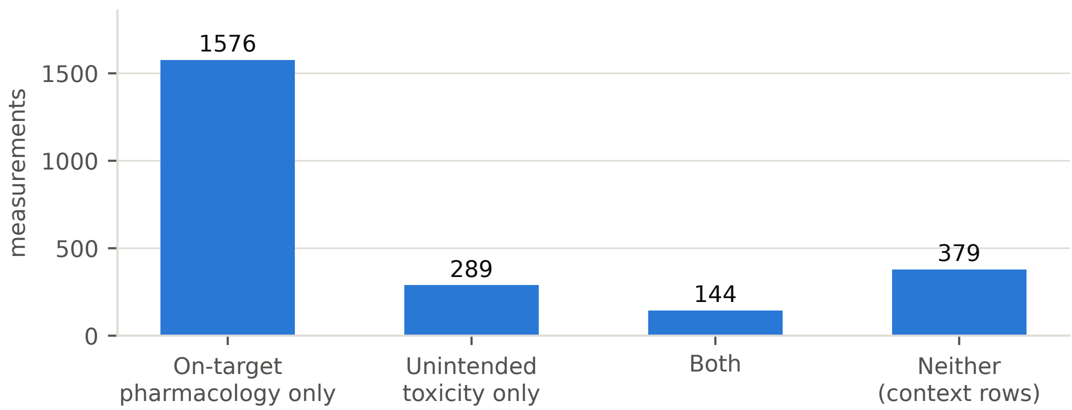

Figure 2. The two axes. On-target pharmacology dominates the row count, which is why the flags exist.

5. The gap this addresses

Coagulopathy is named in the Challenge brief and is, in the published record, described almost entirely at class level. The characterisation the field works from — that phosphorothioate backbones prolong clotting time at high C_{max}, independent of sequence — appears in review and methods literature as a sentence, naming no compound, no dose and no value. What did not exist was a structured table pairing an identified oligonucleotide, its sequence and its per-position chemistry, with a measured coagulation outcome and a citable locus. This release is that table.

It also fills a narrower gap the field has been explicit about: the separation of *on-target anticoagulant pharmacology* from *unintended coagulation toxicity*. Those two live side by side in the literature — often in the same paper — and are routinely pooled. Here they are two columns.

Finally, the release distinguishes human from animal evidence at row level, including for purified-protein systems, where the human/animal question is invisible in a species field. 886 rows (37%) are human or human-derived, 433 of them from human in-vitro and ex-vivo systems — the category the Challenge identifies as of particular interest.

6. Using this to build a predictive model

The four-table design exposes sequence, per-position chemistry and design predictors against graded, per-condition outcomes, with no join required beyond `oligo_id`. A model builder should:

- **Train on the unintended-toxicity class** (289 rows, plus the 144 rows where an on-target compound produced harm), using the 1,576 on-target rows as a mechanistically-explained positive class rather than as toxicity labels.
- **Use the 572 measured nulls as negatives**, and *not* the `NOT_REPORTED` rows, which encode absence of measurement, not absence of effect. The distinction is the difference between a safety model and a model of what has been studied.
- **Filter `grade_caveat`** before treating grade 1 as signal, and prefer `ratio_to_control` — a continuous target — over the ordinal grade.
- **Respect `species_class`.** Train and validate within a system class, or model the human/animal gap explicitly using the 30 compounds that carry both.
- **Exclude `co_administered_agent` rows** from single-agent models.

The most defensible near-term target is not a grade classifier but a regression on aPTT ratio from backbone composition and phosphorothioate count, trained on the unintended class — the one relationship this release now supports with per-compound numbers.

7. Limitations, stated plainly

- **Grades are provisional and mechanical.** No subject-matter expert has reviewed them.
- **The class-effect quantification rests on one source** (42 of 48 rows).
- **No clinical compound has a published sequence** in the sources used, so sequence-to-phenotype modelling is restricted to patent and preclinical compounds.
- **26 rows have an undetermined system origin** — the source never states whether the proteins used were human. They are marked, not guessed.
- **Prothrombotic rows are dominated by one compound family**, so "hypercoagulability" risks being learned as one drug.
- **6 rows are scope-adjacent** — retained as context but marked, because they are not coagulation readouts.
- **Open defects are enumerated**, not summarised away, in the endpoint dossier.